Bingnan Chen

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Github:https://github.com/ChampionNan

EDUCATION

School of Computer Science, USTC Aug 2017- Jul 2021

Undergraduate Degree in Computer Science

Overall GPA: 3.63/4.3, Major GPA : 3.65/4.3

Winter vacation visiting in National University of Singapore (NUS) Jan 2019

- Completed coursework related to artificial intelligence and big data.
- Worked with teammates to produce a business model.

RESEARCH EXPERIENCES

Research on cross-site scripting (XSS) based on taint tracking for Chromium v8 JHU, June 2020 - Sept. 2020

***Advisor*:** Yinzhi Cao, Professor at School of Computer Science, Johns Hopkins University

- ***Overview:***

  Our goal is to record all listener functions along each data flow, and use them as constraints in a constraint solver. The solver yields a solution given all the constraints, and we can use the solution to generate exploits. This gives us relevant information about the exploit by testing different websites, and we can then use this information to suggest possible methods to reduce web application vulnerability.
- ***1. Record listener functions***
- ***2. Put into constraint solver***
- ***3. Generate exploit and get related information***
- ***4. Come up with effective strategies***

Yoda: A Video analytic pipeline Benchmark University of Chicago, June 2020 - Aug. 2020

***Advisor*:** Junchen Jiang, Professor at School of Computer Science, The University of Chicago

- ***Overview:***

  Our goal is to implement a benchmark to understand how different types of video content features affect video analytic pipelines performance, to understand relative strengths and weaknesses of two video analytic pipelines under a range of settings and efficiently estimate performance of video analytic pipelines on new videos.
- ***1. Implement video analytic pipeline interface & profile & performance estimator***
- ***2. Put forward new attempts and improvements to the pipeline***

User Activity recognition and Identification through Wireless Backscattering USTC, Nov 2019- Jan. 2020

***Advisor*:** Panlong Yang, Professor at School of Computer Science, USTC

- ***Overview:***

  Our goal was to achieve accurate and non-intrusive user activity recognition and then user identification through battery-free wireless backscattering.
- ***1. Signal Preprocessing***
- ***2. Adaptive Activity Detection and Segmentation***
  - Ran preliminary activity detection with the help of short-time energy (STE).
  - Use a heuristic segmentation algorithm called BG.
- ***3. Deep Learning Based Motion Recognition and User Identification***
  - Use three modules, namely: feature extraction, attention and parallel classification modules to perform motion recognition and user identification, leveraging the extracted spectrogram.

PROJECT EXPERIENCES

C-- Compiler | Supervisor: Prof. Cheng Li Sept. 2019 - Dec. 2019

https://github.com/ChampionNan/C-minus-compiler

- ***Overview:***

  There are three main steps to finish a C-- compiler: lexical analyzer, syntax analyzer, intermediate code generation, and

RISC-V code production.

- ***1. Lexical analyzer:***

  Used flex to build a lexical analyzer, running analyses and outputting tuples of five attributes for syntax analysis.

- ***2. Syntax analyzer:***

  Used bison to build a syntax analyzer, depending on the input tokens to do certain actions and producing a syntax tree.

- ***3. Intermediate code generation:***

  Used LLVM C++ interface to generate LLVM IR, exchanging syntax trees generated in previous steps to C++ syntax trees with the help of visitor mode.

Summer exchange in Shanghai Jiao Tong University (SJTU) June 2019

https://github.com/ChampionNan/AI-interaction-in-sjtu

- Built a translation machine that recognizes user voice and produces the translation.
- Integrated application of visual and audio information, realizing facial recognition, you-language learning, augmented image processing, human face swapping, and position recognition.

Bank Management system | Supervisor:Prof. Peiquan Jin June 2020

https://github.com/ChampionNan/Bank_Management_System

- ***Overview:*** The front end of the project uses Vue, and the back end uses Flask plus Mysql to implement a bank management system with customer management, account management loan management, and business statistics functions.

Online Interview system | Supervisor:Prof. Jing Li June 2020 - Sept. 2020

https://github.com/ChampionNan/Online_Interview_System

- ***Overview:*** The project uses Vue as the front-end and Django as the backend to build an online user interview platform including collaborative whiteboard, code editor, chat dialog, etc.

RISCV CPU | Supervisor:Prof. Xuehai Zhou May 2020

- ***Overview:*** Implemented RISC_V_32I CPU, group-connected and fully-connected cache (FIFO, LRU), branch prediction part (BTB, BHT). -- language: Verilog and System Verilog.

SCHOLARSHIPS

- Outstanding Student Scholarship (Bronze) 2018

  For top 20% students at USTC

- Outstanding Student Scholarship (Bronze) 2019

  For top 20% students at USTC

- Outstanding Student Scholarship (Freshman) 2017

  For top 20% students at USTC

PROGRAMMING-SLILLS

- ***Languages***: C/C++, Python, MATLAB, Verilog & System Verilog, HTML & CSS & JSP, x86 Assembly language, MySQL
- ***Skills***: LaTeX, TensorFlow, NumPy & Matplotlib, Git, OOP, Django, Vue, Flask

LANGUAGES

- Chinese: Native fluency
- English: TOEFL: 95(R:26, L:25, S:25, W:19)

中国科学技术大学

University of Science and Technology of China

学业成绩表

Scholastic Record of University of Science and Technology of China

Reg.NO: PB17111650 Name: Chen Bingnan Gender: Female Date of Birth: 1998-10-31

Enrl Date: 2017-09-01 Dep Date: E.S: 4

School: Department of Computer Science and Technology

Major: Computer Science and Technology All Curriculum GPA: 3.63

Term	Course Title	Gr.	Hrs.	Cr.	Term	Course Title	Gr	Hrs	Cr.
2017FA	History of The Internet: Technology, Fortune & The Changes of Civilisation	B+	40	2	2018SP	Badminton I	B+	40	1
2017FA	College English Viewing, Listening and Speaking (Band 1)	84	20	1	2018SP	Freshman Seminar	A	20	1
2017FA	Program Design I	83	80	3	2018SP	C Language Programming II	87	80	3
2017FA	Military Theory	Pass	40	1	2018SP	College English Viewing, Listening and Speaking (Band 2)	85	20	1
2017FA	Ideological and Moral Cultivation and Basics of Law	87	60	3	2018SP	Fundamentals of Marxism	86	60	3
2017FA	Basic Sports	B+	40	1	2018SP	Electromagnetism C	86	60	3
2017FA	Mechanics and Thermal Physics	87	80	4	2018SP	College Physics Experiment I	83	60	1.5
2017FA	College English for Reading & Writing (Elementary)	90	40	2	2018SP	College English for Reading & Writing (Intermediate)	85	40	2
2017FA	Introduction to Computer Science	Pass	20	1	2018SP	Algebraic Structure	78	60	3
2017FA	Calculus for Functions of a Single Variable	89	120	6	2018SP	Linear Algebra B1	85	80	4
2018SP	Principles of Economics	A-	60	3	2018SP	Calculus for Functions of Several Variables	85	120	6
2018FA	Table Tennis II	B+	40	1	2019SP	Mathematical Logic	90	60	3
2018FA	College English Viewing, Listening and Speaking (Band 3)	82	20	1	2019SP	College English Viewing, Listening and Speaking (Band 4)	81	20	1
2018FA	Introduction to Computer Systems(H)	87	100	4	2019SP	Probability and Statistics B	87	60	3
2018FA	An Outline of Modern and Contemporary Chinese History	84	40	2	2019SP	Introduction to Chinese Important Thoughts	83	60	3
2018FA	College Physics Experiment II	83	60	1.5	2019SP	Equations of Mathematical Physics B	96	40	2
2018FA	Optics and Atomic Physics	89	80	4	2019SP	Operations Research	73	40	2
2018FA	Syllabus of Digital Logic Lab	85	40	1	2019SP	Operating System	90	100	4
2018FA	Analog and Digital Circuits	78	80	4	2019SP	Computer Organization	90	100	4
2018FA	Graph Theory	86	60	3	2019SP	Computer System	A-	40	2
2018FA	Function of Complex Variable B	92	40	2	2019SU	Selective Course	Pass	40	2
2019FA	Microcomputer Principles and Systems B	91	70	2.5	2019FA	Computer Networks	84	80	3.5
2019FA	Badminton II	A-	40	1	2019FA	Data Structures	96	100	4
2019FA	Foundations of Algorithms	83	90	3.5	2019FA	Principles and Techniques of Compiler	89	100	4
2019FA	Stochastic Processes B	97	40	2					

Note: (H) represents the curriculum of Honors; FA:Fall SP:Spring SU:Summer

GPA Calculation:

Centesimal Grade:	100~95	94~90	89~85	84~82	81~78	77~75	74~72	71~68	67~65	64	63~61	60
Letter Grade:	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-
Point Value:	4.3	4	3.7	3.3	3	2.7	2.3	2	1.7	1.5	1.3	1

$$GPA = \frac{\Sigma(\text{Course Credit} * \text{Course GP})}{\Sigma\text{Course Credit}}$$

Registrar: Dean of School: Guangzhong Sun Undergraduate Education Office: Print date: 2020-02-05 10:33:16

查看成绩

CHEN BINGNAN

ETS Registration Number	3144711190876938
ETS ID	73D9966A
Date of Birth	1998-10-31
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Most Recent Scores :

Test	Test Date	Reading (0-30)	Listening (0-30)	Speaking (0-30)	Writing (0-30)	Total (0-120)
TOEFL iBT	Sun Nov 03 09:00:00 EST 2019	26	25	25	19	95

MyBest™ Scores as of November 07, 2019

Your highest section scores from all valid test dates are shown below.

Test	Reading (0-30)	Listening (0-30)	Speaking (0-30)	Writing (0-30)	Total (0-120)
TOEFL iBT	**26** Test Date: Nov 03, 2019	**26** Test Date: Aug 24, 2019	**25** Test Date: Nov 03, 2019	**21** Test Date: Aug 24, 2019	**98**

How to interpret scores

THE UNIVERSITY OF HONG KONG
Application for Admission to the
Doctor of Philosophy (PhD) and Master of Philosophy (MPhil) Programme
in the Department of Computer Science

Research Statement

Applicant's name:
Bingnan Chen

Proposed Topic/Title of Research:
Improvement of frame selection strategy in real-time object recognition-based video processing system

Background:

The real-time object recognition-based video processing system has emerged due to the universality of the application. Seeking better to reduce network/compute costs without sacrificing accuracy by combining DNN optimization with content-dependent heuristic such as image downsizing and frame sampling, some improving methods about video processing need to be proposed.

From traffic control, factory monitoring to pedestrian tracking, application of live video analytics have recently sprung up in enterprises and smart cities alike. For example, analysis of road-side traffic videos can help detect and respond to vehicular collisions in real time while other initiatives aim to make streets safer with the use of real-time video analytics. Real-time object recognition applications need cameras of good quality located in public area to select and transform video frames as triggered frames by utilizing some heuristic ways and the triggered frames will be leveraged to locate objects and label them with information associated with the objects. Apparently, the client side (camera) has weak computing performance, while the server side will do most of the computationally intensive work. Therefore, how to select a suitable frame to be transmitted with a certain importance on the client is a very important part of the client. There are also many selection methods proposed in many existing pipelines, but most of them are based on heuristics, which means that the performance of good heuristic selections will have a significant impact on the content of the video. My goal is to propose a more universal and accurate frame selection algorithm than the previous heuristic algorithm.

Methodology:

The frame selection proposed here is based on the low-level features about the

frame and the more accurate differences about object in the frame. The previous implementation is to calculate the difference between the pixel points of the current frame and the last transmitted frame. When this difference exceeds a certain threshold, the frame is targeted to transfer. The reason for this is that some parts of a frame belong to a constant background in the animation. Using a more accurate non-background part obviously seems to be more conducive to be closer to the ground truth. In order to achieve such selection operations, we need to deploy a simple neural network on the client side for object recognition. And in order to minimize the omission of object identification, when a certain object from one frame, among a series of consecutive frames, disappears and continues to appear in the frames before and after it, the missing part of this frame will be labeled by comparing the previous and next frames. Such treatment also makes sense, because it may be an oversight in the identification.

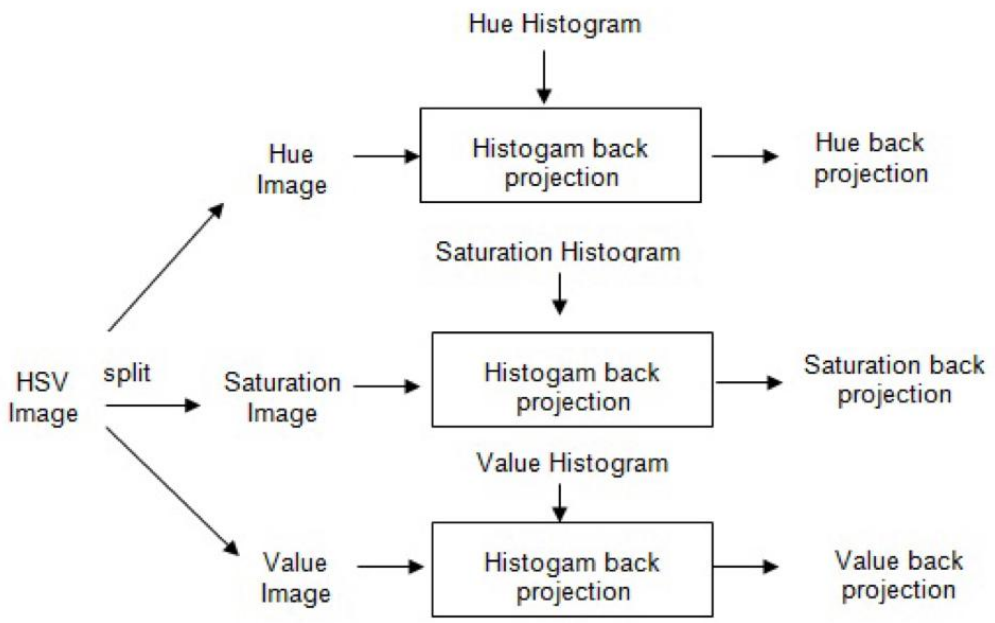

Figure 1. Separate the three components of HSV image

For the part of frame difference, in some cases, the neural network may not perform well because of the limited scale of the training set and direct pixel judgment does not seem to be able to reflect the local characteristics of the image. Therefore, we can also apply other methods for this part to focus on low level feature, mainly HSV (Hue, Saturation, Value) and LBP (Local Binary Pattern) feature about the frame. We will perform following these steps. Firstly, frame conversion where RGB frame is converted into HSV frame and LBP frame. Secondly, utilize histogram back projection, and then separate each HSV image into the hue image, the saturation image and the value image. After that, do histogram back projection to these images. Thirdly, three back projection images are combined into one image using AND operator. What's more, mask LBP Image, which means masking LBP image that we get from image conversion step using

back projection image from the histogram back projection step. This process is carried out by checking each pixel's value in the back projection image. If pixel's value is less than the threshold, then the pixel's value in LBP image that have the same position with the checked pixel is changed into "0". In our experiment, we set the threshold to "255". The threshold is used to eliminate the pixel that has a low probability as object's part. The result of this step is the input image for similarity measure step. As for the measure of similarity, I intend to use Bhattacharyya coefficient.

Further Discussion:

For the implementation of the foregoing operations, there may be a phenomenon that the video correlation is relatively obvious, and there may be a good output on some videos, but this may not be the same case on other videos. We need to find and pay attention to the similarities and differences of these videos, and make preparations for future performance improvements. Also note that the optimization I proposed is based on static video, that is, fixed the client's camera. I don't know if the dynamic performance remains the same.

Future Research Vision:

For the future, what I want to do is related to networks, systems, and security and privacy, then I can be able to do some integration with AI. Because I have always been interested in these directions, I feel that there are many things that need to be exploited and explored in terms of network, system and security. Plus, I think security has an impact on many aspects of our lives, from property security to personal information on the Internet. After doing some network security work, I discovered that there are so many leaked string flows in the network, which contain a lot of private website browsing traces. This phenomenon makes me feel that network security work needs our urgent attention. In terms of AI, I think it is a promising direction. I have taken some related courses and done some related experiments before, especially, I am amazed with its powerful self-learning, self-organization and self-adaptability. It can make a good realization based on some characteristics, which is of great help to our work to some extent. Therefore, if I can combine them and do some scientific research, that would be very exciting.

References:

[1] Tiffany Yu-Han Chen, Lenin Ravindranath, et al. Glimpse: Continuous, Real-Time Object Recognition on Mobile Devices, SenSys'15, November 1-4, 2015, Seoul, South Korea.
[2] Uzma Noreen, Mutiullah Jamil, et al. Hand Detection using HSV Model, Computer Engeering and Intelligent Systems, Vol.6, No.12, 2015
[3] Priyanto Hidayatullah, Politeknik Negeri Bandung. Color-Texture Based

Object Tracking Using HSV Color Space and Local Binary Pattern, International Journal on Electrical Engineering and Informatics, June, 2015

Statement of Purpose

When I was in elementary school, I used Adobe Flash to make a simple animation of a small turtle for a computer operations class. Since then, I have been fascinated by the unique framework that computers utilize to understand the world and their ability to explain or analyze seemingly chaotic systems. After majoring in computer science at the University of Science and Technology of China (USTC), I've realized that learning about computer science is particularly rewarding because it requires extensive mathematical, programming, and analytical skills. Eventually, my passion for the subject has led me to pursue future studies and, ultimately, a career in computer science.

My major as an undergraduate at USTC is in computer science and technology. Among the courses I have studied, courses relating to mathematics and systems have been prominent, such as Operating System (90/100), Microcomputer Principles and Systems B (91/100) , Stochastic Processes (97/100), Equations of Mathematical Physics B (96/100) and other quantitative courses.

Since sophomore year, I've participated in three research projects, based on signal processing, network security & privacy, and video analytics pipelines. My first foray into computer science research started in the signal area combined with neural networks, designed for tasks in user identification. In this project, I developed capabilities for exploring real project scene through signal acquisition, signal processing, and clever application of neural networks. In my work, I mainly tried a new angle of action test and learned how to collect and process data using signal denoising and signal cutting. Finally, I sent the cut data to the next steps to generate spectrograms for subsequent signal recognition. This experience helped me develop a strong interest in scientific research.

During this summer, I started my second project, based on network security regarding cross-site scripting (XSS) vulnerabilities. In order to explore the field of network security, I decided to contribute to the research led by Prof. **Yinzhi Cao** (JHU). Our work is based on the previous work of a published paper from Network and Distributed Systems Security (NDSS), where the method of confirming the attack was proposed. However, their method of confirming DOM XSS attacks was relatively simple and straightforward, and we proposed a novel and more sophisticated method of confirming attacks. I followed the method in the paper to find the taint flow, extracted all the details of the flow (string operation), and then put them as constraints into the constraint solver to find more exploitable flows. Specifically, I made the sink contains these attack vectors, and then leveraged the constraint solver to solve the corresponding string value in the source. Then, assigned this value to the source and ran Chromium, an open sourced version of Chrome. If the web page displays a message about an attack vector, it means that success has absorbed an exploit. In this way, I can identify the value of the source

that cannot be filtered out by the sanitizer and can also trigger the exploit – not found by the previous method – which explained why our method was theoretically more effective. At this point, I have generated the proper flows and enough details for the constraint solver. My next step is to crawl a lot of websites for testing later. In my exploration of this area, I feel that my interest and experience has been expanded very well. After that, I will need to conduct more in-depth studies in the field of network security to better continue this work.

While conducting research about security & privacy, I also participated in some projects working on video analytics pipelines, led by Prof. **Junchen Jiang** (UChicago). The work focuses on implementing an interface for videostorm, a pipeline proposed in usenix symposium on networked systems design and implementation, and awstream, a pipeline proposed in sigcomm. I summarized the video pruning methods into three categories: temporal pruning, spatial pruning and model pruning. Different pipelines can be implemented in this mode, and the encapsulation of the pipeline is well completed, offering the benefit of portability to users. This work in video analytics has helped me develop good understanding of this field and an interest in learning. At the same time, it has improved my ability to build project systems.

During winter and summer vacations, I also participated in NUS programs and SJTU exchange project, where I made a business model and a smart speaker based on Python face recognition on the Raspberry Pi. Moreover, I also gained a lot of experience with AI & Big Data, which improved my general computer skills. I am very interested in personal English speech and I participated in the school's English speech contest where I won the silver award.

The University of Hong Kong is distinguished for its eminent professors, individualized nature, and value of intercultural communication. I would like to reiterate that I have sufficient background, competence, and the motivation to pursue a PhD at this institution. I hope you will regard my admission favorably, and I look forward to embracing a promising future at HKU from next fall.
Thank you for your time and consideration.